



Manufactured opinions: The effect of manipulating online product reviews

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ABSTRACT

Previous research assumes that consumers can detect and discount the manipulation of online product reviews or are oblivious to such practices. We posit that the equilibrium occurs due to the cues of manipulation, consumer suspicion and their expertise. Our analysis of hotel occupancy data shows that the effect of adding positive reviews and deleting negative reviews on sales exhibits an inverted U-curve. Moreover, weak brands suffer more from excessive adding. Our laboratory experiments find that adding affects consumer purchase intention, but it also arouses suspicion, which exerts a negative mediating effect. Deleting is more disguised and difficult to be suspected. Novices are more influenced by manipulations compared with their experienced counterparts. Thus, contrary to the popular belief of “fake it until you make it,” excessive adding leads to consumer distrust and may backfire. Deleting exacerbates information asymmetry and results in adverse selection, thus warrants restraint and regulation.

1. Introduction

Given the tremendous impact of online product reviews on consumer purchases, firms may manipulate online reviews to increase sales by anonymously posting favorable reviews and/or deleting negative reviews (Hu, Bose, Gao, and Liu, 2011). As a half-star difference in a product rating can make or break a business (Anderson & Magruder, 2012), “reputation management” has become a fast-growing industry (Morrison, 2011; Northrup, 2009). By some estimates, up to one-third of all “consumer” reviews on the Internet are fake (Hu, Bose, et al., 2011; Jindal & Liu, 2008). Despite the efforts by e-marketplace operators and review platforms to filter out fake reviews and the strengthening of guidelines and enforcement in various countries, manipulation of online reviews has persisted and taken more varied forms.

Game-theoretical researchers suggest that when rival firms monotonically increase their manipulations of online reviews regarding product quality, online recommendations remain credible given a sufficiently large quantity of consumer reviews (Mayzlin, 2006). In addition, manipulations may increase the informativeness of online reviews for firms with better-quality products despite the noise in the system (Dellarocas, 2006). These studies assume that consumers are aware that manipulation takes place and can update their beliefs and adjust their interpretations of online opinions accordingly. However, research on opinion spam shows that consumers may not be able to detect manipulations or correct for the related bias (Hu, Liu, and Sambamurthy, 2011). As more firms join this “arms race” of manipulation (Dellarocas,

2006), managers often find themselves in a prisoner's dilemma (Gössling, Hall, & Andersson, 2018).

Due to the covert nature of review manipulation, few studies have assessed its effect on sales. Thus, there is little empirical evidence on the benefits of manipulation for firms or the economic effect of such “misinformation.” In addition, while studies have examined the posting of fake positive reviews (e.g., Lappas, Sabnis, & Valkanas, 2016; Luca & Zervas, 2016; Mayzlin, Dover, & Chevalier, 2014), the practice of deleting or hiding negative reviews—another prevalent but more disguised form of review manipulation—remains unexamined in the literature. More importantly, the literature offers little guidance to managers because it has not explored the theoretical mechanism underlying the effect of review manipulation. Given these research gaps, several important questions need to be addressed: 1) How does review manipulation affect product sales? 2) Are the effects of adding fake positive reviews different from those of deleting or hiding negative reviews? 3) Does manipulation result in adverse selection by consumers, where better-quality products are outsold by those with poor brand reputation? 4) To what extent can consumers detect manipulation and discount the manipulated reviews?

We believe that the answers to these questions lie in both the availability of manipulation cues and the ability of consumers to detect and adjust for manipulated reviews. First, drawing from research on deception detection and consumer psychology, we propose that intensive manipulations inevitably leave traces and dampen consumer confidence and product sales. Thus, we posit that review manipulation,

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both adding and deleting, exhibits an inverted U-curve relationship with product sales, in that manipulation initially increases sales, but excessive manipulation is counter-productive. We also examine how the effect of manipulation differs across firms with various levels of brand strength.

Second, compared with persuasion via advertising and promotions, covert manipulations of online reviews are complex, disguised, and more difficult for ordinary consumers to detect. To assess the ability of consumers to suspect manipulation and adjust their purchase intention, we propose that manipulations arouse consumer suspicion, which has a negative mediating effect on purchase intention. Moreover, as adding fake positive reviews and deleting or hiding negative reviews differ in the availability of manipulation cues, these practices arouse suspicion and affect consumer purchase intention to different degrees. We further propose that consumer expertise in online shopping moderates the effect of manipulation on consumer suspicion.

Given the dearth of evidence on the effect of review manipulation, this is the first study to empirically assess its effect on product sales using both field data and a laboratory experiment, and it makes several important contributions to the literature. While research based on the efficient market assumption suggests that an equilibrium in manipulation occurs due to competition among firms and consumers' awareness of the manipulation (Dellarocas, 2006; Mayzlin, 2006), thinkers of the new economics emphasize information asymmetry and lament the vulnerability of consumers (Akerlof & Shiller, 2015). To bridge the gap between these two streams of literature, we believe that studies of consumer psychology and information processing can shed light on the mechanism underlying the effect of review manipulation. Drawing from research on deception detection and consumer persuasion knowledge, our study focuses on the role of cue availability and consumer suspicion as the key underlying mechanism in this process. Our results on the effects of review manipulation—through both adding and deleting—and on the roles played by firm brand strength and consumer suspicion and expertise help to further understanding of the impact of this growing problem in Internet retailing and yield significant theoretical insights and managerial implications.

The following sections are organized as follows. We first provide a succinct review of the literature on online product reviews and seller manipulations. Then, drawing from research on deception detection and consumer psychology, we articulate our research framework and hypotheses. Next, we present the results from our analysis of field data and two laboratory experiments. Lastly, we discuss the findings and their implications for consumer welfare, e-commerce, and public policy, and explore the directions for future research.

2. Literature review

2.1. Online product reviews

Increasing numbers of marketing scholars have highlighted the effects of online product reviews, including in the areas of sales and marketing strategies (e.g., Chen, Wang, & Xie, 2011; Cui, Lui, & Guo, 2012) and consumer decisions (Sen & Lerman, 2007; Smith, Menon, & Sivakumar, 2005). Others have examined the value of online reviews for sales forecasting (Dellarocas, Zhang, & Awad, 2007) and consumer motivations for posting reviews (Chen et al., 2011). Researchers have used a number of metrics to examine the effect of online word-of-mouth (WOM), including the volume of reviews, average ratings (valence), and the distribution of ratings (dispersion).

The literature suggests that (1) the quality and reliability of reviews are critical factors, and (2) the volume and valence of reviews (i.e., average rating) have a significant effect on product sales (e.g., Cui et al., 2012; Smith et al., 2005). Overall, studies have found that the valence of reviews has varying degrees of influence on consumer purchases and sales, with the standardized beta up to 0.33 for various products, including movies, books, games, and consumer electronics (e.g., Cui

et al., 2012; Dellarocas et al., 2007). Researchers have also found that due to the negativity bias among consumers, the negative effect of negative reviews on sales is greater than the positive effect of favorable reviews (e.g., Cui et al., 2012). As a result, firms have strong incentives to manipulate reviews to influence consumer perceptions by altering both the quantity (i.e., the number of reviews) and quality (rating and content) of the reviews (Northrup, 2009).

2.2. Manipulating online reviews

In addition to advertising and promotions, consumers increasingly rely on online reviews to assess the quality of products before making purchase decisions. The reviews posted by previous buyers are considered an unbiased reflection of product quality. To influence consumer perceptions, sellers may manipulate the reviews of their products on company websites, review platforms, and e-marketplaces (e.g., TripAdvisor and Amazon). Recently, operators of e-marketplaces and online forums such as Amazon and Yelp have installed opinion spam algorithms to filter out fake reviewers and their reviews. Businesses have also been prosecuted for such fraudulent and illegal practices, which have non-trivial implications for consumer welfare and the development of e-commerce.

Numerous reports suggest that the manipulation of online reviews by skills is a widespread and growing practice. For example, book publishers and sellers offer various incentives for favorable reviews, such as \$25 gift cards for positive reviews of textbooks (Northrup, 2009). Multinational companies such as Bayer, Levi's, Starwood Hotels, Mazda, KFC, and Kraft have all used buzz techniques to influence consumers' purchase decisions (Mayzlin, 2006). It has been reported that because fake fan accounts now sell faster than stolen credit card numbers, the Zeus virus software used to steal credit card data has been modified to create bogus Instagram "likes" to generate buzz for companies, with 1000 "likes" going for \$30 (Finkle, 2013). Aside from adding positive reviews, sellers also hide or delete negative reviews. In exchange for a monthly fee, Yelp allegedly offered to move a firm's negative reviews to the bottom of its Yelp page and make it difficult for users to access them (Morrison, 2011). Clearly, adding positive reviews is different from deleting negative reviews in terms of costs and benefits.

Scholars have explored how manipulation of online product reviews affects firms and consumers. Mayzlin (2006) proposes an analytical model in which two competing firms supply anonymous messages praising their own products, and suggests that despite the distortions from deceptive messages, online WOM remains informative given a sufficient amount of authentic reviews. Dellarocas (2006) describes the conditions in which the number of manipulated reviews increases with the quality of the firms and concludes that manipulations from high-quality firms potentially lead to more informed customer decisions. These studies suggest that although manipulations may decrease the informational value of online reviews, they may not affect consumer decisions because consumers are aware that manipulation takes place and sophisticated enough to discount and adjust their interpretations of online opinions.

Due to the covert nature of review manipulation, direct evidence of such practices is lacking. Nevertheless, researchers have used various methods to detect and quantify the extent of manipulation in online reviews (Hu, Bose, et al., 2011; Hu, Bose, Koh, & Liu, 2012). Hu, Liu, and Sambamurthy (2011) focus on the detection of fake positive reviews using a sophisticated classification algorithm and text analysis. Jindal and Liu (2008) propose using "sentiment analysis" to filter out the authors of fake reviews. These researchers suggest that the complex and disguised nature of review manipulation makes it difficult for ordinary consumers to detect or adjust for the bias. Mayzlin et al. (2014) estimate the extent of review manipulation based on a comparison of reviews on Expedia and TripAdvisor. They find that an increase in incentives to manipulate reviews leads to a greater amount of

manipulation, although the level of manipulation is small relative to the total number of reviews. Moreover, independent, less reputable businesses are more likely to manipulate reviews (Luca & Zervas, 2016; Mayzlin et al., 2014). Their results suggest that in contrast to the advertising literature, small firms of inferior quality engage in more manipulation activities because they have more to gain. Fake reviews can be used to increase a hotel's visibility or to attack its competitors (Lappas et al., 2016; Mayzlin et al., 2014). Thus, firms vary in their motivations to manipulate online reviews and in the benefits they may derive from such activities.

From consumers' perspective, recent studies using consumer surveys and laboratory experiments find that informational and normative cues significantly affect consumers' adoption of online reviews (Filiari, 2015) and that manipulation dampens information quality and trustworthiness, reducing the effectiveness of reviews (Reimer & Benkenstein, 2016). Despite the different propositions, real data on review manipulation are typically not available, making it a challenge to estimate its effect on product sales. To date, there is neither empirical evidence nor a coherent theoretical framework on how review manipulation influences sales. Finally, while researchers have focused on the posting of fake positive reviews (Mayzlin, 2006), they have not examined the practice of deleting or hiding negative reviews and how different manipulation tactics affect sales and consumer perceptions.

3. Conceptual framework and hypotheses

Scholars of information economics suggest that information asymmetry is prevalent in e-commerce, especially for complex products, and that consumers, being not perfectly informed, may not make optimal or rational decisions (Stiglitz, 2002). In a recent book, Akerlof and Shiller (2015) argue that in “a phishing equilibrium,” firms manipulate information to influence consumers for extra profit, and competition alone does not eliminate manipulation but may exacerbate the situation. In an increasingly competitive market, firms may find engaging in manipulation the most rational choice (Gössling et al., 2018). Game-theoretical studies, however, suggest that both the cost of manipulation and the ability of consumers to detect and adjust for the manipulated reviews lead to an equilibrium in manipulation (Dellarocas, 2006; Mayzlin, 2006). This suggests that the self-interest of firms should serve as a self-correcting “natural law” to minimize the pitfalls of manipulation. Clearly, the difference between the marginal cost and benefit of manipulation affects a firm's actions.

We believe that both the cost of manipulation and consumer responses are critical for answering the research questions. First, aside from the reputational risk of being caught engaging in such activities (Mayzlin et al., 2014), we posit that the cost of manufacturing fake reviews is not constant but rather increases over time as firms engage in more manipulations. Although firms may selectively release or hide information to their own advantage, manipulated reviews invariably leave cues of deception and reduce the value of the reviews. Thus, manipulation is not without pitfalls. When done excessively, manipulation is more difficult and costly and easier to detect, leading to consumer suspicion and discontent. The degree to which consumers suspect manipulation depends on the availability of cues associated with these tactics. Second, given the increased cues of manipulation over time, consumers have more opportunities to process these cues and become suspicious. Although consumers may or may not detect the manipulation activities, they can gain important knowledge about online reviews from their own experience and other sources (Ahlwalia & Burnkrant, 2004). Thus, the persuasion knowledge of consumers may limit the effect of manipulation to some extent. Based on these rationales and drawing from research on deception detection and the persuasion knowledge model, we propose the following conceptual framework (Fig. 1) and elaborate the research hypotheses.

3.1. Adding fake positive reviews

Previous game-theoretical models suggest that if posting fake positive reviews monotonically provides more information about a product's quality and if there is a sufficient number of reviews, the overall review system remains persuasive to consumers and may even become more informative (Dellarocas, 2006; Mayzlin, 2006). Thus, manipulation of online reviews can help to sell more products, or at least have a non-negative effect on product sales. Moreover, these models assume that consumers are aware that such manipulation takes place and may discount the information, resulting in an equilibrium of manipulation among firms. Although firms may engage in optimal levels of review manipulation to avoid being caught, both consumers and firms benefit more if firms manipulate less (Dellarocas, 2006).

However, this does not mean that adding fake positive reviews has a monotonic linear relationship with product sales. There are several reasons to believe that the effect may instead exhibit an inverted-U shape. First, the informational value of online reviews primarily stems from authentic customers, who are perceived to have greater warranting value and to reflect the true quality of products, thus facilitating comparison shopping. The warranting value of information is defined as the extent to which it is perceived as immune to manipulation, and it varies depending on whether the information is generated anonymously or can be verified as authentic (Walther, Van Der Heide, Hamel, & Shulman, 2009). In online forums, warrants are the cues that an observer uses to gauge the authenticity and value of product reviews, such as verified IDs and the requirement of a purchase for posting reviews, which make the reviews more trustworthy. Websites that do not have these fact-checking mechanisms are prone to manipulation and are more likely to have abnormal distributions of positive and negative reviews that deviate from consumer expectations.

Secondly, adding one or two reviews does not make a difference to the review system and may not have the desired effect. It is reasonable to assume that some sellers continually add fake reviews. However, while it may be relatively easy to add a few fake reviews and avoid detection, the more manipulation firms engage in, the more difficult it is for them to write fake reviews without being suspected. Excessive adding inevitably leaves more traces, leading to suspicion and discounting of the reviews (Anderson & Simester, 2014) while also decreasing the informational value of reviews and increasing the reputational risk to firms (Mayzlin et al., 2014). In other words, when reviews are manipulated to the point that they arouse consumer suspicion, the product quality becomes questionable. When consumers suspect fake reviews, they pay more attention to the consistency and veracity of the information. As a result, consumers understand that products with possible fake reviews are of lower quality and adjust their perceptions accordingly.

Third, studies of persuasion strategies in advertising and communication have found an inverted U-curve relationship between the intensity of a strategy and its effect on consumers (Shu & Carlson, 2014; Teixeira, Picard, & Kaliouby, 2014). Kirmani (1990) finds that the perceived cost of advertising affects brand perception in an inverted U-curve manner, in which consumers consider marketers to be manipulative when the perceived cost of advertising is high. Thus, the intensity of a persuasive strategy tends to have a positive effect on consumer response before it levels off and then becomes negative. As sellers treat the posting of fake reviews as a promotional strategy or persuasion attempt, we expect such practices to have a similar effect.

Based on the above discussion, we posit that the effect of adding fake reviews on sales is a convex function of the manipulation effort or intensity. Adding fake reviews in an attempt to increase the number of positive reviews and affect consumer perceptions of product quality is expected to have an initially positive effect on product sales. However, as adding fake reviews becomes more frequent and intensive, there comes a point when the effect wears away and consumer suspicions are aroused, leading to a detrimental effect on sales thereafter.

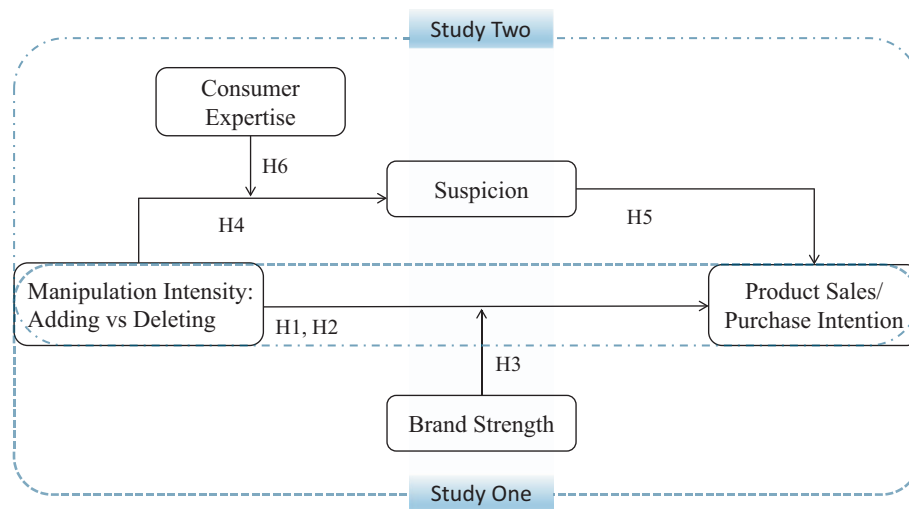


Fig. 1. The conceptual framework.

Hypothesis 1. Adding fake positive reviews has a positive effect on product sales up to a certain point before the effect begins to turn negative in the manner of an inverted U-curve.

3.2. Hiding or deleting negative reviews

In addition to the review content, manipulation tactics differ in terms of the availability of cues, level of disguise, and ease of detection. Adding involves presenting inauthentic information, typically in the form of “fake” reviews by people who have not made a purchase. Adding may involve the use of questionable customer IDs, extreme positive statements, and skewed rating distribution (Hu, Bose, et al., 2011). Adding as “cheap talk” is more likely to leave cues of manipulation and arouse suspicion. Consumers with sufficient experience may be able to observe the anomalies in such reviews.

Although hiding or deleting negative reviews does not involve posting false information, it changes the valence and distribution of reviews. Using various reputation management tools such as search engine optimization, firms can push negative reviews to the second page or even further (Quipp, 2008). By definition, deleting negative reviews can be classified as lying or deception by omission (Fulmer, Barry, & Long, 2009). Because the hidden or deleted messages are typically not visible to readers, there are fewer cues available for deleting in comparison with adding (Hu, Bose, et al., 2011). Thus, hiding or deleting is more disguised and less likely to be suspected. Even deleting is not without pitfalls, however, given that consumers specifically seek out negative reviews due to the negativity bias, i.e., the greater value that consumers attach to negative reviews (e.g., Cui et al., 2012). Given the importance of negative reviews to consumers, we propose the following hypothesis:

Hypothesis 2. Deleting negative reviews has a positive effect on product sales up to a certain point before the effect begins to turn negative in the manner of an inverted U-curve.

3.3. Brand strength

Mayzlin (2006) and Dellarocas (2006) suggest that manipulation is more valuable for a firm whose product is less popular with the target segment than for a firm whose product is more appealing. A possible explanation is that firms with less reputable products spend more on advertising compared with firms with popular products, as the latter firms believe that consumers are likely to receive information from other sources praising their superior products (Moorthy & Hawkins,

2005). Thus, reviews of products with low average ratings are more likely to be manipulated (Hu, Bose, et al., 2011). Ho-Dac, Carson, and Moore (2013) find that positive (negative) reviews increase (decrease) the sales of weak brands but have no significant effect on the sales of strong brands. Mayzlin et al. (2014) find that compared with large chain hotels, smaller independent hotels are more likely to manipulate online reviews because they have more to gain. Therefore, firms with weak brands may engage in more review manipulation and expect to gain from such practices because there are (1) not enough positive reviews for less popular products, and/or (2) too many negative reviews for low-quality products.

Moreover, firms with strong brands are less influenced by the negative effect of intensive manipulation. First, as explained above, firms with strong brands are less likely to manipulate online reviews than their counterparts with weak brands. Second, consumers have access to many sources of information to assess popular brands, such as brand reputation (e.g., official star rankings) and advertising. Finally, in the event of negative publicity, studies show that consumers are more forgiving of big brands, as their reputation provides a certain amount of protection (Einwiller, Fedorikhin, Johnson, & Kamins, 2006). Pullig, Netemeyer, and Biswas (2006) suggest that when consumers have a high degree of certainty about their prior brand attitudes, they may counter-argue and dismiss the negative publicity, whereas less favorable attitudes toward a brand may aggravate the effect of negative publicity. Therefore, we hypothesize that weak brands are more susceptible to the negative effects associated with excessive manipulation.

Hypothesis 3. Excessive adding (3a) and deleting (3b) have a less negative effect on product sales of strong brands in comparison with weak brands.

3.4. Suspicion

Studies of review manipulation typically use secondary data and have not examined the effect of manipulation on consumer suspicion and purchase intention. Thus, whether consumers suspect manipulative intent and adjust their evaluations of reviews remains unexplored. Studies of consumer information processing suggest that observed deviation from expectations or norms leads to consumer suspicion of manipulative intent, and the content and manner of the communication help consumers to assess the salience of such deviation (Ahluwalia & Burnkrant, 2004). Thus, the availability of manipulative cues is essential for assessing such deviation and arousing consumer suspicion. Studies in advertising and personal selling assume that the cues are accessible to consumers (Friestad & Wright, 1994). However,

consumers in computer-mediated online settings typically do not have access to the same cues to detect deceptions (Anderson & Simester, 2014).

Nevertheless, the content and features of online reviews offer clues for consumers to assess their authenticity. This is especially true for adding as opposed to deleting, as the manipulation tactics are often embedded in the message content or rhetorical style, time and sequence of posting, and questionable reviewer IDs and names through which paid shills try to conceal their identities (Hu, Bose, et al., 2011). As manipulators alter the content of messages one at a time, adding fake positive reviews leaves more cues of manipulation, thus arousing stronger consumer suspicion. In contrast, deleting negative reviews leaves fewer cues for readers, although the small number or absence of negative reviews may arouse suspicion.

Hypothesis 4. Adding fake positive reviews leads to greater suspicion than deleting negative reviews.

Research on information manipulation suggests that increased suspicion enhances the emotional intensity in situations where either the lie or the act of lying is considered to be significant (McCornack & Levine, 1990). Studies of persuasion also show that consumers' suspicion of manipulative intent may disrupt the comprehension and elaboration of a message, and that the lack of trust and confidence leads consumers to discount the recommendation (Campbell & Kirmani, 2000; Friestad & Wright, 1994). Although manipulations may alter consumer perceptions and increase the likelihood of a purchase, the suspicion aroused by manipulations may decrease the purchase possibility. When consumers suspect manipulation in reviews, they are likely to disengage from the purchase process, draw inferences about the authenticity of reviews, and discount the recommendations of the reviewers and the product quality (Dellarocas, 2006; Mayzlin, 2006). As a result, suspicion of manipulation dampens confidence in reviews and products and leads consumers to adjust their interpretations of online reviews, resulting in lower purchase intention. Thus, while manipulation may affect consumer purchase intention, it also leads to suspicion, negatively mediating the effect of manipulation on purchase intention.

Hypothesis 5. Suspicion mediates the effect of adding (5a) and deleting (5b) on purchase intention.

3.5. Consumer expertise

While the availability of cues raises consumer suspicion, not all consumers are able to perceive the cues of manipulation. Consumer knowledge and expertise play a key role in this process. The persuasion knowledge model suggests that consumers use their knowledge of the motives and tactics of persuasion to interpret, evaluate, and respond to the persuasive attempts of marketers and others (Campbell & Kirmani, 2000; Friestad & Wright, 1994). The level of persuasion knowledge depends on the consumers' experience and expertise in information searching and other consumption-related activities (Kirmani & Zhu, 2007). When problematic reviews contain obvious abnormalities that deviate from the experience and expectations of consumers, review manipulation can evoke persuasion knowledge. For instance, experienced consumers may suspect adding tactics when encountering extreme positive ratings and remarks that greatly exceed their expectations. Similarly, deleting tactics may be suspected when there is a skewed distribution of ratings or a lack of negative reviews.

Therefore, consumers' online shopping expertise can affect their processing of reviews, perceptions of products, and purchase intention. Experienced consumers are more vigilant about manipulative intent (Kirmani & Zhu, 2007). Apparently, consumers with experience reading online product reviews are more likely to activate their persuasion knowledge, observe these deviations, and raise their suspicion (Ahlualwalia & Burnkrant, 2004; Darke & Ritchie, 2007). Compared with novice online shoppers, consumers with online shopping expertise are

more able to deliberately process online product reviews, filter out abnormal reviews, discount suspicious reviews, and adjust their reliance on reviews—abilities that make them less influenced by manipulation.

Hypothesis 6. Consumer expertise moderates the effect of suspicion on purchase intention in that, compared with consumers with little online shopping expertise, experienced consumers are less influenced by manipulation of reviews through adding (H6a) and deleting (H6b).

Given the elaborate framework and number of hypotheses, it is difficult to obtain the data of all of the variables from a single source. Thus, we conduct two studies to examine the effects of online review manipulation. Study one estimates the effect of the intensity of manipulation on product sales (H1 and H2) and the moderating effect of brand strength (H3) using field data from the hotel industry. Because data on consumer suspicion and shopping expertise are not available from online sources, Study two uses laboratory experiments to examine the effect of manipulation tactics on consumers' suspicion (H4), the mediating effect of suspicion (H5), and the moderating effect of online shopping expertise (H6).

4. Study one: a field study

4.1. The data

To test the first three hypotheses, we collected data from two online hotel review websites and obtained the hotel occupancy data from STR, a major supplier of data on the hotel industry worldwide. Expedia is an online travel agency (OTA) that provides hotel booking services. Only consumers who have booked a hotel through the website in the past six months can post a review of a hotel. TripAdvisor is an international open forum that publishes consumer reviews of hotels, and anyone can post a review on the platform. Visitors to TripAdvisor are directed to other OTAs if they find a hotel desirable. Thus, the cost of manipulating reviews on Expedia is high relative to that on TripAdvisor. To test this conjecture, Mayzlin et al. (2014) find that there are significantly fewer review manipulations on Expedia than on TripAdvisor. Following Mayzlin et al. (2014), we take Expedia as the platform with fewer manipulations and TripAdvisor as the platform with more manipulations, and we use the difference-in-differences approach to measure the extent of manipulation.

We collected the hotel review data from Expedia and TripAdvisor for 35 weeks from January 1, 2014 to August 27, 2014. Given the huge quantity of online reviews of hotels at different travel destinations, we chose the city of San Diego, a major metropolitan area and popular destination for both business and leisure travelers. Unlike a small city or resort town, a major city like San Diego has a variety of hotels for different types of travelers to consider. First, we obtained the ratings of all historical reviews on both websites. Second, we obtained the data of hotel characteristics, room rates (i.e., price), and occupancy rates of hotels from STR. The data from STR were appended to the data of hotel reviews to form a panel data set. The final sample contained 5813 weekly observations of 167 hotels. Table 1 provides the descriptive statistics.

4.2. Measurements

4.2.1. Manipulation intensity

To assess the effect of review manipulations on product sales, we adopt the difference-in-differences (DID) approach to develop a proxy measure of manipulation intensity. Because the average valence is around 4 (3.950 on TripAdvisor vs. 3.828 on Expedia), we consider 5-star reviews as positive, 4- and 3-star reviews as neutral, and 2- and 1-star reviews as negative (Mayzlin et al., 2014). For each hotel, we assume that neutral reviews (i.e., 3- and 4-star reviews) are intact on both websites. We calculate the ratio of positive (negative) reviews against

Table 1
Descriptive statistics of hotels and reviews (study 1).

Variables	Mean	SD	Correlation matrix					
			1	2	3	4	5	6
1. Occupancy	0.789	0.147	1.000					
2. Average room rate	127.176	54.180	0.409***	1.000				
3. Review volume	469.895	491.683	0.161***	0.486***	1.000			
4. Review valence	3.828	0.541	0.216***	0.500***	0.262***	1.000		
5. MIPos	0.005	0.123	0.042	0.234***	0.056	−0.099**	1.000	
6. MINeg	0.046	0.093	−0.104***	−0.088*	−0.029	−0.461***	0.068	1.000

Note: It is not uncommon to have a positive price-demand relationship in certain product/market segment and time period. The relationship can be upward when a market segment is less price-sensitive than the average population, and when demand is strong for premium brands of limited supply, especially during peak seasons.

*** $p \leq .01$.

** $p \leq .05$.

* $p \leq .10$.

the number of neutral and positive (negative) reviews. We use the difference in this ratio between TripAdvisor and Expedia to represent the amount of manipulation and examine its effect on sales. Specifically, for hotel i at time t , we measure the manipulation intensity of positive reviews (hereafter MIPos) and negative reviews (hereafter MINeg) as¹:

$$MIPos_{it} = \frac{5StarReviews^{TA}}{3 + 4 + 5StarReviews^{TA}} - \frac{5StarReviews^{Exp}}{3 + 4 + 5StarReviews^{Exp}} \quad (1)$$

$$MINeg_{it} = \frac{1 + 2StarReviews^{Exp}}{1 + 2 + 3 + 4StarReviews^{Exp}} - \frac{1 + 2StarReviews^{TA}}{1 + 2 + 3 + 4StarReviews^{TA}} \quad (2)$$

The rationale for the measurements is twofold. First, the costs of posting positive reviews are different between the two websites. While anyone can post a review on TripAdvisor without a booking or stay, Expedia only allows verified customers to post reviews. Second, the possibility of deleting negative reviews differs between the two websites. There have been several reports of TripAdvisor removing unfavorable reviews, while such reports are rarer for Expedia.²

The validity of this measure rests on the assumption that the difference in the distributions of hotel ratings varies between the two sites is due to the amount of manipulation (Mayzlin et al., 2014). This measure serves as a good proxy of manipulation intensity for the following reasons: 1) there should be little difference in the proportions of hotel ratings between the two websites given a sufficiently large number of reviews; 2) adding may increase the number of positive reviews, while deleting decreases the number of negative reviews; and (3) the difference in the proportion of positive and negative reviews between the two websites should be divided by the number of neutral reviews on each site. Table 2 compares the review data of the two websites.

A potential source of the differences between the two websites could be the difference in customer preferences due to self-selection. To ensure that key measurements are not influenced by such preferences, we compare reviews of the same hotels between Expedia and TripAdvisor and find that both the ratings and review volumes between the two websites are strongly correlated ($r = 0.784$ and $r = 0.721$, respectively, $p < .001$). These findings suggest that customer preferences are not systematically different between the two websites. Moreover, the characteristics of the reviews (valence and volume, etc.) and hotels (i.e., ownership and reputation, as specified below) are the independent

¹ We assume that deleting or hiding negative reviews is more likely to occur on TripAdvisor than on Expedia; therefore, Expedia should have a greater number of negative reviews.

² After 2015, we found 6 news reports of suspected review manipulation on TripAdvisor and only 1 article regarding Expedia. <http://tripadvisor-warning.com/tripadvisor-accepts-bribes>.

Table 2
Review comparison between TripAdvisor.com and Expedia.com (study 1).

	TripAdvisor		Expedia		t-Test	
Review volume	781.58	(756.20)	469.9	(491.68)	38.81	$p < .01$
Review valence	3.95	(0.49)	3.83	(0.54)	28.85	$p < .01$
% of 5-star reviews	38.92%	(17.46%)	37.15%	(17.35%)	11.59	$p < .01$
% of 4-star reviews	34.03%	(7.62%)	31.90%	(8.23%)	19.21	$p < .01$
% of 3-star reviews	14.83%	(7.59%)	15.39%	(7.17%)	−6.49	$p < .01$
% of 2-star reviews	7.59%	(6.01%)	7.67%	(4.87%)	−1.26	$p > .10$
% of 1-star reviews	4.63%	(5.13%)	7.88%	(9.37%)	−37.82	$p < .01$

Note: Standard errors in parentheses.

variables in the regression models. The hotel characteristics, which may account for consumer preferences, are controlled for, thus minimizing the potential bias from self-selection (Mayzlin et al., 2014).

4.2.2. Brand strength

We operationalize brand strength using both the type of ownership, which indicates the size of the parent firm, if any, and the brand reputation, which is based on the hotel class obtained from STR. Chain hotels are usually big firms with multiple units and well-known brands, while independent hotels are mostly smaller with lesser known brands. Premium hotels are luxury premises rated as three stars and above because of favorable perceptions of their brands and consumer experiences with their facilities and services. Economy hotels are those rated as two stars and below. Based on these two dimensions, we classify the hotels into four categories: 1) chain premium, 2) independent premium, 3) chain economy, and 4) independent economy. Among the four types of hotels, chain premium hotels are considered to have the strongest brands, followed in descending order by independent premium, chain economy, and independent economy hotels.

4.3. The empirical model

We postulate that the occupancy rate is affected by hotel heterogeneities, both observed and unobserved, in the dataset. To account for the hotel-level heterogeneities, we cast our model in a hierarchical framework with random coefficients. For hotel i at week t , occupancy is modeled as

$$\begin{aligned}
Occupancy_{it} = & \alpha_{0i} + \alpha_{1i}MIPos_{i,t-1} + \alpha_{2i}MIPos_{i,t-1}^2 + \alpha_{3i}MINeg_{i,t-1} \\
& + \alpha_{4i}MINeg_{i,t-1}^2 + \alpha_5Brand_{1i} + \alpha_6Brand_{2i} + \alpha_7Brand_{3i} \\
& + \alpha_8Occupancy_{i,t-1} + \alpha_9Log(Volume_{i,t-1}) + \alpha_{10}Valence_{i,t-1} \\
& + \alpha_{11}Log(AveRate_{i,t-1}) + \delta_t + \varepsilon_{it}
\end{aligned} \quad (3)$$

where $MIPos_{i,t}$ and $MINeg_{i,t}$ are the measures of manipulation intensity for adding positive reviews and deleting negative reviews for hotel i at week t . In Eq. (3), we assume that the occupancy of week t will be affected by the reviews and the manipulations of week $t-1$. $Brand$ is a group of dummy variables: chain premium ($Brand_{1i} = 1$), independent premium ($Brand_{2i} = 1$), chain economy ($Brand_{3i} = 1$), and independent economy as the default group. $Volume_{i,t-1}$ and $Valence_{i,t-1}$ are the number and average rating of reviews on Expedia at week t for hotel i . We use the reviews from Expedia because they reflect the true customer evaluations more accurately. $AveRate_{i,t-1}$ denotes the average room rate (i.e., price) for hotel i at week t . Finally, we include month dummies δ_t to control for the seasonal fixed effects.

We capture the hotel-level unobserved heterogeneity with a random coefficient on the intercept by allowing α_{0i} to vary along its population mean α_0 . In addition, we allow the coefficients of manipulation intensity (both positive and negative) and its quadratic terms to vary among their population means and hotel class. These random coefficients are modeled as follows:

$$\begin{bmatrix} \alpha_{0i} \\ \alpha_{1i} \\ \alpha_{2i} \\ \alpha_{3i} \\ \alpha_{4i} \end{bmatrix} = \begin{bmatrix} \bar{\alpha}_0 \\ \bar{\alpha}_1 \\ \bar{\alpha}_2 \\ \bar{\alpha}_3 \\ \bar{\alpha}_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \\ \alpha_{41} & \alpha_{42} & \alpha_{43} \end{bmatrix} \times \begin{bmatrix} Brand_{1i} \\ Brand_{2i} \\ Brand_{3i} \end{bmatrix} + \begin{bmatrix} \omega_0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} \quad (4)$$

We allow all the unobserved heterogeneities to correlate:

$$(\omega_0, \omega_1, \omega_2, \omega_3, \omega_4) \sim MVN(\mathbf{0}, \Sigma) \quad (5)$$

4.4. Results and discussion

We adopt the restricted maximum likelihood estimator (RMLE) to estimate the empirical model. The RMLE is an effective method adopted by prior studies to estimate the mixed effect models (e.g., Snijders & Bosker, 2012). We report the estimated fixed effects in Table 3 and the estimated heterogeneities in Table 4. In model 1 of Table 3, we test the average effects of all hotels regardless of their class. The effect of $MIPos$ on sales is non-zero and positive before a certain point, after which the effect turns negative ($\alpha_1 = 0.094$ and $\alpha_2 = -1.085$, $p \leq .05$). Hence, Hypothesis 1 is supported. In practice, it is important to know where the turning point occurs. The turning point for $MIPos$ is $|0.094 / (2 \times 1.085)| = 0.043$. Thus, adding positive reviews increases hotel bookings when consumers are not aware of their occurrence, but this effect declines when manipulation intensity reaches a certain level, after which consumer suspicion begins to exert a negative effect on sales.

For deleting negative reviews, the significant coefficient and its quadratic term ($\alpha_3 = 0.689$ and $\alpha_4 = -2.497$, $p \leq .05$) of $MINeg$ support Hypothesis 2. The turning point here is $MINeg^* = |0.689 / (2 \times 2.497)| = 0.138$, which is nearly three times the “optimal” level of adding positive reviews. This is not surprising given that deleting negative reviews is more disguised and less likely to be suspected. The comparison between the turning points of adding and deleting is shown in Fig. 2. Thus, the negative effect of excessive deleting is greater than that of adding. This finding is consistent with the concept of negativity bias in that negative reviews are more valuable to consumers than positive ones (e.g., Cui et al., 2012).

Model 2 of Table 3 shows the effects of manipulations for hotels with different levels of brand strength. The estimated coefficients of $MIPos$, $MINeg$, and their quadratic terms suggest that for independent economy hotels, manipulation intensity retains the inverted-U shaped

Table 3

Estimated results (study 1).

Model/variable	Model 1	Model 2
$MIPos_{i,t-1}$	0.094 (0.031)***	0.355 (0.107)***
$MIPos_{i,t-1}^2$	-1.085 (0.192)***	-2.675 (0.708)***
$MINeg_{i,t-1}$	0.689 (0.093)***	-0.273 (0.262)
$MINeg_{i,t-1}^2$	-2.497 (0.742)***	-1.018 (0.366)***
$Brand_1 \times MIPos_{i,t-1}$		-0.293 (0.120)**
$Brand_1 \times MIPos_{i,t-1}^2$		1.632 (0.760)**
$Brand_1 \times MINeg_{i,t-1}$		1.155 (0.292)***
$Brand_1 \times MINeg_{i,t-1}^2$		-1.961 (2.631)
$Brand_2 \times MIPos_{i,t-1}$		-0.144 (0.136)
$Brand_2 \times MIPos_{i,t-1}^2$		-0.237 (1.053)
$Brand_2 \times MINeg_{i,t-1}$		0.712 (0.540)
$Brand_2 \times MINeg_{i,t-1}^2$		-0.878 (3.846)
$Brand_3 \times MIPos_{i,t-1}$		-0.270 (0.123)**
$Brand_3 \times MIPos_{i,t-1}^2$		1.703 (0.771)**
$Brand_3 \times MINeg_{i,t-1}$		1.196 (0.300)***
$Brand_3 \times MINeg_{i,t-1}^2$		-2.066 (2.499)
$Brand_{1i}$	0.024 (0.012)**	-0.064 (0.031)
$Brand_{2i}$	0.016 (0.013)	-0.074 (0.033)**
$Brand_{3i}$	0.005 (0.015)	-0.043 (0.036)
$Occupancy_{i,t-1}$	0.074 (0.004)***	0.065 (0.004)***
$log(AverageRate_{i,t-1})$	0.007 (0.006)	0.004 (0.006)
$log(Volume_{i,t-1})$	0.002 (0.006)	0.008 (0.007)
$Valence_{i,t-1}$	0.010 (0.004)***	0.011 (0.004)***
Constant	0.707 (0.014)***	0.783 (0.030)***
AIC	-1987.583	-1990.653
BIC	-1825.533	-1771.408

Notes: 1) Dependent variable: $Occupancy_{it}$; 2) brand1 = chain premium, brand2 = independent premium, brand3 = chain economy, and independent economy as the default; 3) standard errors are in parentheses.

*** $p \leq .01$.

** $p \leq .05$.

Table 4

Estimated unobserved heterogeneity (study 1).

	Std. dev.	Covariance				
		1	2	3	4	5
Model 1						
1. Constant	0.028	0.001	0.001	-0.021	0.007	-0.079
2. MIPos _{t-1}	0.035	0.001	0.001	-0.026	0.009	-0.094
3. MIPos _{t-1} ²	0.742	-0.021	-0.026	0.550	-0.191	2.097
4. MINeg _{t-1}	0.298	0.007	0.009	-0.191	0.089	-1.036
5. MINeg _{t-1} ²	3.518	-0.079	-0.094	2.097	-1.036	12.375
Model 2						
1. Constant	0.030	0.001	0.002	-0.020	0.009	-0.091
2. MIPos _{t-1}	0.064	0.002	0.004	-0.042	0.019	-0.195
3. MIPos _{t-1} ²	0.706	-0.020	-0.042	0.499	-0.177	1.478
4. MINeg _{t-1}	0.305	0.009	0.019	-0.177	0.093	-1.106
5. MINeg _{t-1} ²	3.960	-0.091	-0.195	1.478	-1.106	15.682

effect for only adding ($\bar{\alpha}_1 = 0.355$ and $\bar{\alpha}_2 = -2.675$, $p \leq .05$) but not deleting ($\bar{\alpha}_3 = -0.273$, $p > .10$ and $\bar{\alpha}_4 = -1.018$, $p \leq .05$). To test the interaction between excessive manipulation and brand strength, we examine the coefficient of $brand \times MIPos^2$ and $brand \times MINeg^2$. For chain premium hotels, the interaction of brand strength with the quadratic term of $MIPos$ is significant and positive ($\alpha_{21} = 1.632$, $p \leq .05$), while the interaction with $MINeg^2$ is not. For independent premium hotels, the coefficients of interaction terms regarding $MIPos^2$ and $MINeg^2$ are not significant. For chain economy hotels, the coefficient of interaction with $MIPos^2$ is significant ($\alpha_{23} = 1.703$, $p \leq .05$), while the interaction with $MINeg^2$ is not. Thus, H3b regarding the interaction between brand strength and $MINeg^2$ is not supported. This may be largely due to the disguised nature of deleting negative reviews.

In post hoc analyses to examine the regression slopes of the interactions between brand strength and $MIPos^2$, we re-estimated the model using each type of hotel as the reference group. For H3a on “excessive”

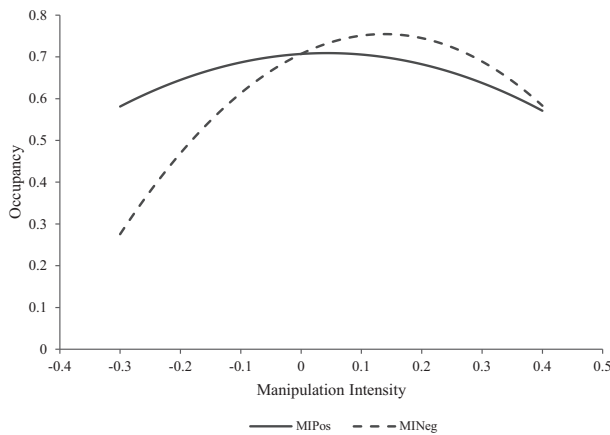


Fig. 2. The effect of manipulation intensity on hotel occupancy (study 1).

adding, we compare the coefficients of interactions of brand strength with the squared terms of MIPos: $\alpha_{\text{chain premium}} = -1.038$, $\alpha_{\text{independent premium}} = -0.917$, $\alpha_{\text{chain economy}} = -2.863$, and $\alpha_{\text{independent economy}} = -2.675$. For chain and independent premium hotels with stronger brands, their coefficients are significantly less negative than those for chain and independent economy hotels ($p \leq .05$), furnishing support for H3a.

5. Study two: laboratory experiments

Although the analyses of field data reveal how the extent of review manipulation may affect product sales, they are not based on direct observations of consumers and thus cannot take into account suspicion and online shopping expertise. To assess the role of consumer suspicion and expertise, we conducted two experiments to test our hypotheses.

5.1. Study 2a

To develop the scenarios for the two manipulation tactics, we conducted in-depth interviews with 16 experienced online shoppers, who discussed the two types of manipulations they had experienced and the cues for suspecting them. The cues and heuristics they mentioned were largely consistent with the ones found by studies of opinion-spam filtering (Hu, Bose, et al., 2011). Based on insights from the interviews and previous studies, the adding scenario involved the webpage of a fictitious hotel with 1) six overwhelmingly positive comments, 2) dubious user IDs, and 3) more positive reviews than the distribution of reviews and ratings (number of stars) would indicate. Deleting contained only positive reviews and no negative reviews, even though the distribution of ratings clearly indicated a number of negative reviews. The control group of intact reviews without any manipulation featured six reviews that were a mix of positive, neutral, and negative reviews with normal reviewer IDs and names. To simulate the effect of manipulation, the average product rating (i.e., 4.0/5.0) was consistent for the two treatment groups, while the average rating of the control group

was a bit lower (i.e., 3.5/5.0). All other information (i.e., hotel name, webpage format, and product information such as price) was kept constant across the three scenarios.

5.1.1. Measurements

We measured purchase intention using a single item: “Supposing the price of this hotel is reasonable and fits your budget, how likely are you to choose this hotel?” Suspicion was measured by a three-item semantic differential scale that rated the online product reviews as “unbelievable/believable,” “not truthful/truthful,” or “deceptive/non-deceptive” (Kirmani & Zhu, 2007). We adapted the three-item scale (i.e., knowledge, experience, and expertise) from Jain and Posavac (2001) to measure online shopping expertise, ranging from very little to a great deal. These variables were all measured on a 7-point scale with a neutral point in the middle (i.e., 4).

5.1.2. Manipulation checks

To validate the scenario designs, we conducted a pre-test and manipulation check for the type of review manipulation (i.e., adding vs. deleting vs. control) by asking the subjects to identify the three scenarios. From a university in Hong Kong, we recruited 20 graduate students who had online shopping experience. They were first briefed about online hotel reviews and the related manipulation tactics, and then asked to read three scenarios and match them with specific types of manipulation tactics. The results showed that all three scenarios were identified and matched correctly (19 out of 20, 95% correct), thus indicating the validity of the scenario designs.

5.1.3. The main study

Two hundred and nine subjects were recruited from the online consumer panel M-Turk to participate in an experiment featuring a hotel review website. The subjects were from different regions of the world and required to have experience in online shopping. The subjects were randomly assigned to one of three scenarios simulating a realistic online hotel booking website, seeing the six most recent reviews on the hotel homepage. The reviews were extracted from TripAdvisor and adapted to the scenarios. They first read a description asking them to book a hotel for an upcoming trip and mentioning a fictitious hotel, Salinas. After the introduction, they were presented with the review webpage in an online format similar to a typical hotel review page. They were instructed to read the reviews as if they were in an information search situation before making their purchase decision. After 90 s, the subjects were presented with the questionnaire and asked to indicate their purchase intention and perception of the hotel. They then completed the second part of the questionnaire to indicate their suspicion of review manipulation, online shopping expertise, and demographic information. The sample comprised 46.5% males and 53.5% females, with ages ranging from 21 and 70 years old.

5.1.4. Results and discussion

Table 5 reports the descriptive statistics of the variables. The Cronbach's alphas of suspicion and online shopping expertise are 0.933 and 0.970, respectively. To test hypothesis 4, we ran two regressions

Table 5
Descriptive statistics of variables in the experiment (study 2a).

Variables	Mean	SD	Alpha	Correlation matrix				
				1	2	3	4	5
1. Purchase intention	5.017	1.265	na	1.000				
2. Suspicion	2.611	1.059	0.933	−0.455***	1.000			
3. Online expertise	5.234	1.052	0.970	0.007	−0.111***	1.000		
4. Gender	1.541	0.499	na	0.037	0.023	−0.070	1.000	
5. Age	34.961	10.087	na	−0.037	0.030	−0.067	0.112***	1.000

*** $p \leq .01$.

Table 6
The Effect of manipulation on suspicion and purchase intention (study 2a).

Dependent variable	Suspicion	Purchase intention
Adding	0.194 (0.082)**	0.420 (0.062)***
Deleting	−0.177 (0.064)***	0.573 (0.068)***
Adding × expertise	0.183 (0.091)**	
Deleting × expertise	0.191 (0.072)***	
Suspicion		−0.401 (0.069)***
Expertise	−0.191 (0.095)**	0.018 (0.054)
Age	0.007 (0.067)	−0.057 (0.047)
Gender	0.025 (0.065)	0.051 (0.052)
R ²	0.138***	0.454***
Adj. R ²	0.108***	0.438***

Notes: 1) Standardized beta; 2) standard errors in parentheses; and 3) expertise is mean-centered.

*** $p \leq .01$.

** $p \leq .05$.

with suspicion and purchase intention as the dependent variables. The results in Table 6 suggest that the effect of adding on suspicion is significant and positive ($\beta = 0.193$, $p \leq .05$) in comparison with the control group, while the effect of deleting on suspicion is significant and negative ($\beta = -0.177$, $p \leq .01$). Thus, deleting is less likely to raise consumers' suspicion than adding, supporting Hypothesis 4. The results also indicate significant interactions between consumer expertise and the type of manipulation ($\beta_{\text{adding}} = 0.183$, $p \leq .05$; $\beta_{\text{deleting}} = 0.191$, $p \leq .01$). The significant positive interaction effects on suspicion suggest that consumer expertise makes them more suspicious of manipulation. Altogether, the main effects and moderating effects reveal that the subjects are more suspicious of adding, and consumer expertise raises their suspicion level only marginally. Deleting negative reviews is more disguised and thus suspected to a much lesser degree, although experienced consumers are more likely to suspect deletion.

Table 6 suggests that suspicion has a significant negative effect on purchase intention ($b = -0.401$, $p \leq .01$). To test the mediating effect of suspicion moderated by online shopping expertise, we ran a regression using the Hayes, Preacher, and Myers (2011) procedure for analyzing the moderated mediation effect with categorical predictor variables. As shown in Table 7, both adding and deleting have a significant positive direct effect on purchase intention ($b_{\text{adding}} = 1.122$ and $b_{\text{deleting}} = 1.519$, $p \leq .01$). The indirect mediating effect on purchase intention in the adding path is significantly negative for consumers with greater expertise ($b_{\text{high}} = -0.422$, $p \leq .01$), but it is not significant for consumers with less expertise ($b_{\text{low}} = 0.007$, *n.s.*). For the deleting path, the indirect effect is not significant for consumers with greater expertise ($b_{\text{high}} = -0.018$, $p > .10$), but it is significant and positive for consumers with less expertise ($b_{\text{low}} = 0.395$, $p \leq .01$). These results suggest that suspicion mediates the relationship between adding and purchase intention for experienced consumers and the relationship between deleting and purchase intention for less experienced consumers, partially supporting H5a and H5b. Again, the results further

reflect the different degrees to which the two manipulation methods are disguised. A simple *t*-test illustrates that adding is rated as the most suspicious (adding: 3.007 vs. control: 2.568, $p \leq .05$) and deleting as the least suspicious (deleting: 2.210 vs. control: 2.568, $p \leq .01$). As adding is less disguised, experienced consumers consider it more suspicious than the intact scenario, while subjects with less expertise fail to do so. In contrast, since deleting is more disguised, high-expertise subjects may view it as intact, while low-expertise subjects consider it more trustworthy than the intact scenario.

As for the moderating effect of online shopping expertise, in the case of adding (Table 7), the negative mediating effect of suspicion for the high-expertise group is greater than that for the low-expertise group ($b_{\text{high}} = -0.422$ vs. $b_{\text{low}} = 0.007$), and the difference between them is significant ($b = -0.429$, $p \leq .05$). As for the deleting path, the mediating effect of suspicion for the high-expertise group is significantly less positive than that for the low-expertise group ($b_{\text{high}} = -0.018$ vs. $b_{\text{low}} = 0.395$), and the difference between them is significant ($b = -0.413$, $p \leq .01$). These results suggest that the mediating role of suspicion is significantly stronger for the high-expertise group than for the low-expertise group in both cases, furnishing support Hypothesis 6.

The total effects in Table 7 indicate that both adding and deleting increase the purchase intention ($b = 0.700$, $b = 1.129$, $b = 1.502$, and $b = 1.914$, $p \leq .01$). The results suggest that hotels may benefit from manipulating their online reviews in the short run. To simulate a realistic review manipulation environment, the valence (i.e., average rating) of the control scenario (intact reviews) is 0.5 lower than that of the two treatment scenarios in this experiment. Thus, the effect of review manipulation on purchase intention may be partially due to the increased valence, aside from heightened suspicion.

5.2. Study 2b

To further tease apart the combined effects of valence and suspicion, we replicate the first experiment using a fixed valence (i.e., with 4.0/5.0 rating) for all three scenarios. Other than that, experiment 2b adopts the same measures of variables, design of scenarios (i.e., number of positive vs. negative reviews, user IDs, and webpage format), and procedures as experiment 2a. Thus, the details are not repeated here.

Subjects were recruited at a major university in Hong Kong to participate in an experiment featuring hotel services. The subjects were asked to imagine that they intended to book a hotel room for an upcoming trip and were presented with a hotel homepage and related product information, followed by six reviews. Then, the subjects were asked to complete the first part of the questionnaire indicating their purchase intention. Following that, they were given the second part of the questionnaire to indicate their suspicion of manipulative intent in the reviews, online shopping expertise, and demographic information. The experiment ended with a usable sample of 180 subjects (33.3% male), with ages ranging from 18 to 44 years old.

The Cronbach's alpha of the two multi-item measures (i.e., expertise

Table 7
The mediating effect of suspicion moderated by online shopping expertise (study 2a).

Mediator: suspicion		Indirect	Direct	Total
Path	Adding → purchase intention			
	High expertise	−0.422*** (−0.829, −0.129)	1.121*** (0.801, 1.430)	0.700*** (0.169, 1.194)
	Low expertise	0.007 (−0.278, 0.297)		1.129*** (0.722, 1.515)
	Difference	−0.429** (−0.947, −0.016)	−	−0.429** (−0.947, −0.016)
Path	Deleting → purchase intention			
	High expertise	−0.018 (−0.222, 0.209)	1.519*** (1.178, 1.917)	1.502*** (1.104, 1.895)
	Low expertise	0.395*** (0.192, 0.669)		1.914*** (1.585, 2.307)
	Difference	−0.413** (−0.765, −0.107)	−	−0.413** (−0.765, −0.107)

Notes: 1) 95% confidence-level in parentheses; 2) the results are generated from 1000-times bootstrapping; 3) the control group is the reference; and 4) expertise is mean-centered.

*** $p \leq .01$.

** $p \leq .05$.

Table 8
Estimated results (study 2b).

Dependent variable	Suspicion	Purchase intention
Adding	0.174 (0.081)**	0.147 (0.080)*
Deleting	0.027 (0.078)	0.347 (0.079)***
Adding × expertise	0.219 (0.091)**	
Deleting × expertise	0.201 (0.098)**	
Suspicion		−0.310 (0.071)***
Expertise	0.007 (0.110)	−0.056 (0.073)
Age	0.052 (0.068)	0.036 (0.069)
Gender	0.093 (0.072)	−0.002 (0.084)
R ²	0.132***	0.207***
Adj. R ²	0.096***	0.180***

Notes: 1) Standardized beta; 2) standard errors are in parentheses; and 3) expertise is mean-centered.

*** $p \leq .01$.

** $p \leq .05$.

* $p \leq .10$.

and suspicion) are 0.867 and 0.917, indicating good reliability. We follow the same estimation procedures used in study 2a, and report the main results in Table 8 and the path analysis in Table 9. Table 8 shows that the estimation results are largely consistent with those of study 2a, except that the direct effects of review manipulations on purchase intention have disappeared (adding: $\beta = -0.071$, $p > .10$; deleting: $\beta = 0.117$, $p > .10$). The results suggest that review manipulations have no effect on purchase intention if the review ratings are not affected by the manipulation. In other words, review manipulation affects consumers' purchase intention through the increased review ratings. The total effects in Table 9 further reveal that if the ratings are fixed, high-expertise consumers are more likely to be suspicious of the questionable reviews than low-expertise consumers. In contrast to adding, deleting is less likely to backfire. Specifically, adding dampens the purchase intention of high-expertise consumers ($b = -0.480$, $p \leq .05$) but not that of the low-expertise group ($b = -0.172$, $p > .10$). Deleting marginally increases the purchase intention of low-expertise consumers ($b = 0.427$, $p \leq .10$) but not that of the high-expertise group ($b = 0.136$, $p > .10$).

Thus, study 2b confirms that suspicion toward reviews plays a mediating role between review manipulation and purchase intention, and that deleting negative reviews, as a more disguised form of manipulation, is less suspicious than adding fake positive reviews. In addition, study 2b also confirms that consumer expertise moderates the effect of suspicion, with experienced consumers more likely to be suspicious of questionable reviews.

Table 9
Path analysis results (study 2b).

Mediator: suspicion		Indirect	Direct	Total
Path	Adding → purchase intention			
	High expertise	−0.295*** (−0.586, −0.103)	−0.185 (−0.581, 0.203)	−0.480** (−0.887, −0.088)
	Low expertise	0.012 (−0.211, 0.180)		−0.172 (−0.593, 0.263)
	Difference	−0.308*** (−0.686, −0.072)	−	−0.308*** (−0.686, −0.072)
Path	Deleting → purchase intention			
	High expertise	−0.167 (−0.445, 0.030)	0.303 (−0.102, 0.704)	0.136 (−0.363, 0.574)
	Low expertise	0.124 (−0.064, 0.344)		0.427* (−0.028, 0.902)
	Difference	−0.291** (−0.720, −0.024)	−	−0.291** (−0.720, −0.024)

Notes: 1) 95% confidence-interval in parentheses; 2) the results are generated from 1000-times bootstrapping; 3) all scenarios display 4-star rating; and 4) expertise is mean-centered.

*** $p \leq .01$.

** $p \leq .05$.

* $p \leq .10$.

6. Conclusions

6.1. Findings and contributions

The results from the field study suggest that although both types of review manipulation have short-term benefits, excessive manipulation leaves more cues of alteration, raises consumer suspicion, and eventually leads to a negative effect on product performance. The study also shows that weak brands are more likely to suffer from the negative effect of excessive adding. Thus, the effects of review manipulation are asymmetric across firms. The results of two laboratory experiments reinforce and extend the findings of the field study in that, while manipulations may lead to increased purchase intention (in terms of both direct and total effects), the negative mediating role of suspicion is significant for posting fake positive reviews, but not for deleting, which is more disguised. As fewer cues of manipulation are available in the case of deleting, consumers cannot readily suspect manipulative intent, thus resulting in higher purchase intention and adverse selection. However, consumers with substantial expertise in online shopping are less subject to the influence of manipulation due to their heightened suspicion. Consumers with less online shopping experience may fall victim to manipulation activities.

This study is the first to empirically assess the effect of online review manipulation on product sales and consumer purchase intention, and it contributes to the literature in several important ways. While the game-theoretical research suggests that, given a sufficient quantity of real reviews, an equilibrium in manipulation occurs due to consumers' awareness of manipulation (Dellarocas, 2006; Mayzlin, 2006), we draw from the research on deception detection and consumer psychology and highlight the role of cue availability and consumer suspicion. These two factors serve as the missing link in the existing literature, shed light on the mechanism underlying the non-linear effect of review manipulation, and highlight its pitfalls. In addition, our studies examine the effect of deception by omission, i.e., deleting or hiding negative reviews. These two manipulation tactics—i.e., adding and deleting—differ greatly in terms of the availability of manipulation cues and their effect on firms and consumers. The effect of excessive adding varies across firms with different levels of brand strength, but the same is not true of deleting, which is more disguised. Moreover, our research suggests that consumers are neither fully aware of manipulations so as to readily discount them nor completely oblivious to such activities. Whether they can detect manipulations and adjust their perceptions depends on the availability of cues associated with manipulation tactics and their expertise in online shopping.

6.2. Implications

Overall, the results from the two studies provide strong evidence that manipulating online product reviews has a non-negligible and

complex effect on product sales and consumer purchase intention. The results enrich our understanding of this critical issue and have meaningful implications for e-commerce operators, industry development, and public policy. For businesses that post fake positive reviews to get ahead of the competition, our findings suggest that although these manipulation activities have a short-term benefit, they eventually arouse consumer suspicion and dampen their confidence. Contrary to the widely held belief of “fake it until you make it,” these results suggest that excessive manipulation is counterproductive and self-defeating, especially for less reputable businesses. As fewer cues are available for deleted negative reviews, consumers with limited online shopping experience will not readily suspect the manipulations and may fall prey to such practices. However, even deleting is not without pitfalls, as experienced consumers may seek out negative reviews due to the negativity bias.

For firms caught up in the “arms race” of manipulating reviews in responding to their competitors, our findings suggest that they need not be overly concerned with the undue advantages of manipulation by others. Our findings highlight the pitfalls of manipulating online reviews. Adding fake reviews may make products more attractive, but it also eventually arouses consumer suspicion. Drawing on their persuasion knowledge, consumers may suspect manipulative intent and discount the product reviews. Thus, firms and consumers are better off when there is less or no manipulation. Despite the “cheap talk” and noise in review systems, the open forums on the Internet remain a great equalizer, as honest firms with better products eventually prevail. Product quality/brand popularity and uncontaminated public opinion are still the arbitrators of the e-marketplace. Thus, instead of squandering time and effort in the fruitless game of “manufacturing opinions,” marketing resources can be better spent on enhancing brand reputation and winning the trust of consumers. Managers should continually monitor customer feedback and public opinion to learn about customer preferences, respond to praise and criticism proactively, and make online reviews more valuable for them.

These findings highlight a number of critical issues that deserve the attention of industry associations and public policy makers. Online consumer reviews have become a new tool for consumer information search and decisions. Manipulations of online reviews, especially deleting or hiding negative reviews, are deceptive and unethical practices that have a significant effect on consumer welfare. Unsuspecting consumers may be affected by manipulators who, camouflaged as real users, attempt to shore up the ratings of their products or drown out disapproving voices. If unscrupulous sellers use a review platform for phishing, the “collective filtering” system may diminish in value. The resulting lack of trust may lead consumers to place greater weight on negative reviews and/or seek opinions from other channels, resulting in generalized suspicion among consumers. Manufactured opinions, if not regulated, can be detrimental to the development of e-tailing, leading to vicious cycles of consumer discontent and mudslinging among competitors.

Industries and regulators need to develop consistent guidelines for distinguishing legitimate “reputation management” from fraudulent practices. If unchecked, these practices lead to greater suspicion among consumers, reduce the informational value of online product reviews, and may render them useless. Manipulative activities should be monitored and contained to ensure the sustainability and healthy development of e-commerce. The community should develop policies and provide technical support for eliminating manipulations. While opinion spam filters may help identify the manipulators and filter out their fake messages, deleting negative reviews is less noticeable, exacerbates the information asymmetry for less experienced consumers, and is detrimental to consumer welfare. Thus, the suppression of negative reviews should be monitored and made impossible or costly. To make reviews helpful to consumers and valuable for firms, aside from installing opinion spam filters, website operators should require reviewers to have verified user IDs and purchases to uphold the authenticity and veracity

of the reviews.

6.3. Limitations and future research

We conducted a field study and two laboratory experiments to examine the effect of manipulations on product sales and consumer purchase intention, and our methodology and findings are of broad interest. However, readers may note that this research has several limitations. Both the field study and the experiments use only one product category. Thus, caution should be exercised when generalizing these findings to other product categories and populations. As the detection of manipulations requires rich and confidential data, future researchers may consider collecting data from other companies and product categories. Given the scope of the study and the difficulty of classification, the case of posting negative reviews about competitors was not considered in this study. Due to privacy restrictions, background characteristics of reviewers are not available, but these characteristics may reflect unobservable heterogeneity and affect their ratings of hotels. Future studies may examine other measures of consumer perception and behavior. For example, behavioral measures can go beyond purchase intention to include attitude change in terms of the perceived helpfulness of online reviews and trust in online sellers. Finally, researchers may explore how firms, instead of deleting or ignoring negative reviews, can proactively respond to unfavorable feedback, gain the respect of consumers, and win their trust.

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